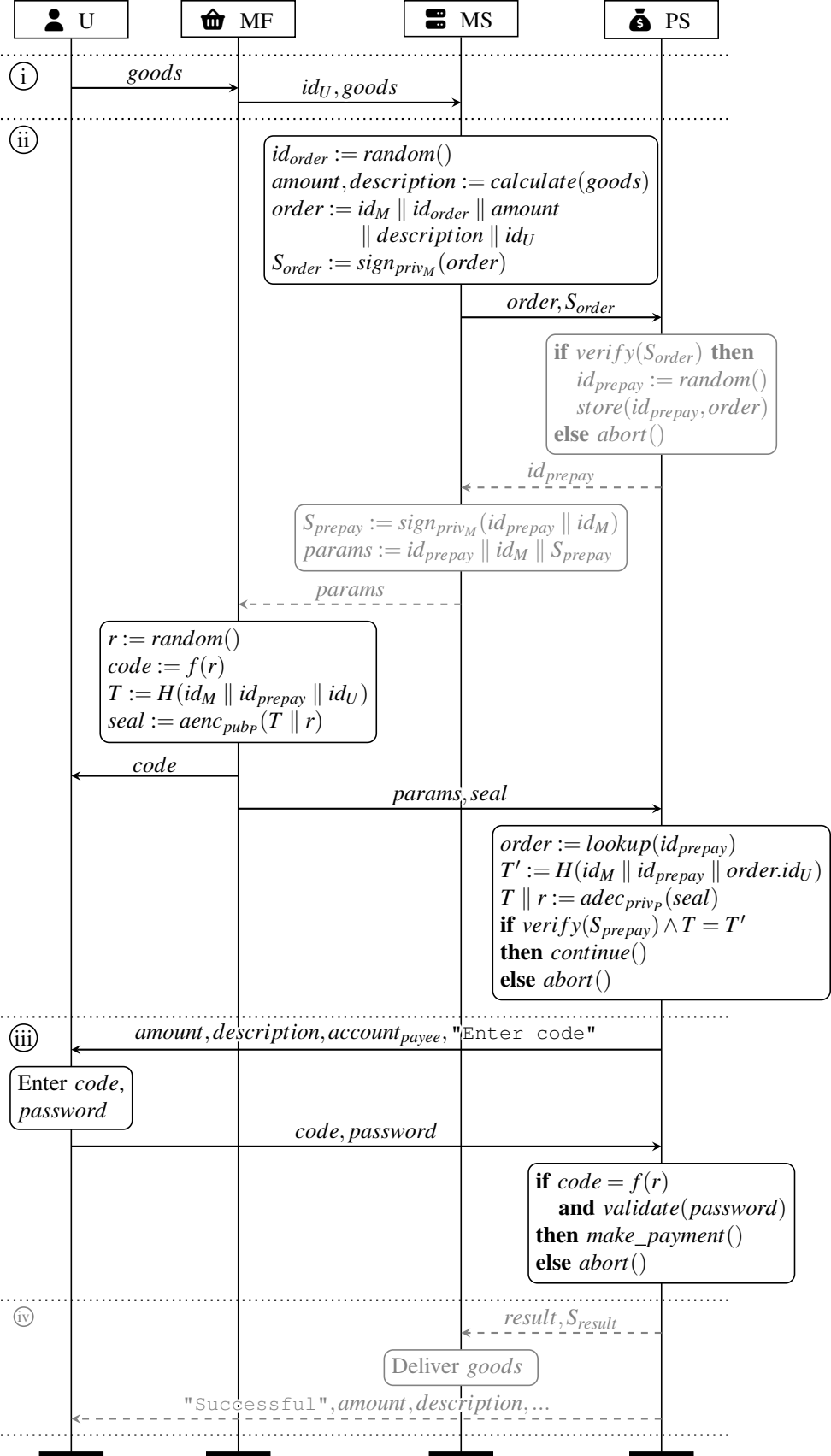

U
MF
MS
PS
i
goods
$id_U, goods$
ii
$id_{order} := random()$
$amount, description := calculate(goods)$
$order := id_M \parallel id_{order} \parallel amount \parallel description \parallel id_U$
$S_{order} := sign_{priv_M}(order)$
$order, S_{order}$
**if** $verify(S_{order})$ **then**
$id_{prepay} := random()$
$store(id_{prepay}, order)$
**else** $abort()$
$id_{prepay}$
$S_{prepay} := sign_{priv_M}(id_{prepay} \parallel id_M)$
$params := id_{prepay} \parallel id_M \parallel S_{prepay}$
$params$
$r := random()$
$code := f(r)$
$T := H(id_M \parallel id_{prepay} \parallel id_U)$
$seal := aenc_{pub_P}(T \parallel r)$
$code$
$params, seal$
$order := lookup(id_{prepay})$
$T' := H(id_M \parallel id_{prepay} \parallel order.id_U)$
$T \parallel r := adec_{priv_P}(seal)$
**if** $verify(S_{prepay}) \wedge T = T'$
**then** $continue()$
**else** $abort()$
iii
$amount, description, account_{payee}$, "Enter code"
Enter *code*, *password*
$code, password$
**if** $code = f(r)$
**and** $validate(password)$
**then** $make_payment()$
**else** $abort()$
iv
$result, S_{result}$
Deliver *goods*
"Successful", $amount, description, ...$